



TWO-FRAME SURVEY TELEMETRY

Defect ID: 5aa6cd35-3ed6-4f82-8982-b17d4b0bfcda

Coordinates: 12.840610, 80.153408

Current Scan: 2026-09-08 10:00:00 UTC

Time Delta: 7d 0h ago

Two-Frame Spatial-Temporal Comparative Metrics

CURRENT DEFECT AREA

0.6500 m<sup>2</sup>

Prev: 0.5000 m<sup>2</sup>

AREA DELTA %

+30.0%

State: Worsening

MUNICIPAL PRIORITY

Rank 8 / 10

Severity: High

Civil Engineering Advisory (Gemini 1.5 Flash)

Root Cause Analysis

Asphalt fatigue cracking aggravated by subsurface water seepage.

Recommended Remediation

Full-depth asphalt patching and edge drainage installation.

Pavement Condition Index (PCI) Technical Appendix

Standardized ASTM D6433 road survey assessment methodology applied. Structural integrity reflects localized shear stress and cyclic asphalt thermal contraction. Municipal action schedule requires hot-pour rubberized mastic application within 14 operational days to mitigate base-course gravel dispersion and water ingress.

Quality Assurance & Municipal Compliance

Survey verified through automated multi-modal visual vector comparison and RTK-calibrated GPS waypoint logging. All reported surface deformations adhere to AASHTO Highway Maintenance Guidelines for localized distress mitigation.



# ScanTrail Engineering Certification & Audit

Municipal Pavement Condition & Verification Record

## AASHTO / ASTM Compliance & Maintenance Matrix

| Parameter                    | Measured Value                       | Threshold / Tolerance             |
|------------------------------|--------------------------------------|-----------------------------------|
| Defect Identifier            | 5aa6cd35-3ed6-4f82-8982-b17d4b0bfcd4 | Unique UUID-4 Spatial Key         |
| Geodetic Coordinate (Lat)    | 12.840610°                           | WGS-84 RTK < 0.05m tolerance      |
| Geodetic Coordinate (Lon)    | 80.153408°                           | WGS-84 RTK < 0.05m tolerance      |
| Surface Area Delta           | +30.00%                              | Evolution Threshold: ±15.0%       |
| Current Bounding Geometry    | 0.6500 m²                            | Calibrated Orthomosaic Grid       |
| Previous Bounding Geometry   | 0.5                                  | Baseline Historical Scan          |
| Temporal Audit Interval      | 7d 0h ago                            | Dynamic Rover Visit Cycle         |
| Municipal Priority Scale     | 8 / 10                               | 1 (Restored) to 10 (Critical)     |
| Target Work-Order Completion | 14 Calendar Days                     | AASHTO Standard Maintenance Cycle |

## Municipal Engineer Sign-off & Verification Statement

This document certifies that the automated road survey conducted at the specified geodetic coordinates has been logged into the municipal spatial asset register. The multi-modal matching engine confirmed the temporal defect progression via two-frame vector alignment. The actionable recommendations provided are approved for procurement and contractor dispatch under standard municipal infrastructure maintenance protocol.

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Authorized Municipal Road Inspector

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ScanTrail Autonomous Telemetry Engine